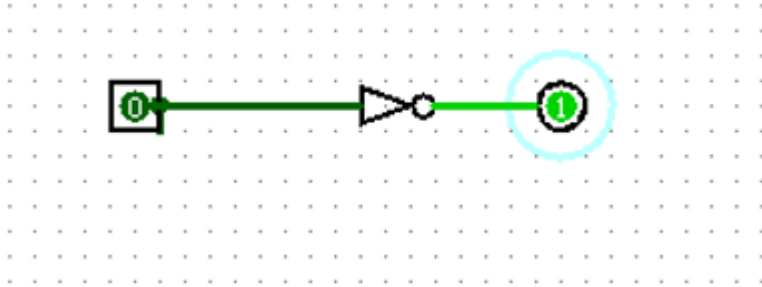


1) Verification of basics logic gates: AND, NOT and OR:

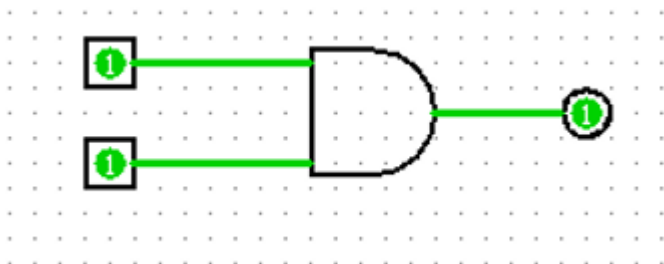
- **NOT gate (Inverter):** Outputs the *opposite* of the input.



Truth table

a	x
0	1
1	0

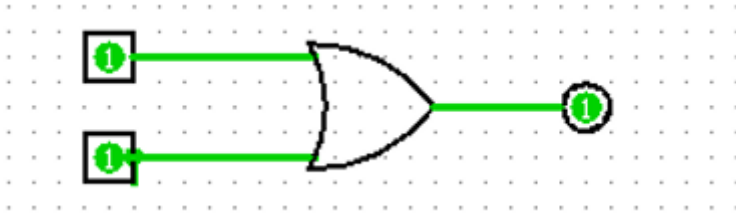
- **AND gate:** Outputs 1 only if *all inputs are 1*.



Truth table

a	b	x
0	0	0
0	1	0
1	0	0
1	1	1

- **OR gate:** Outputs 1 if *any input is 1*.

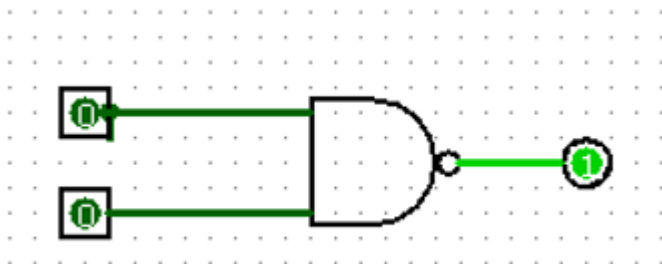


Truth table

a	b	x
0	0	0
0	1	1
1	0	1
1	1	1

2) Verification of derived logic gates: NAND, NOR, XOR, XNOR:

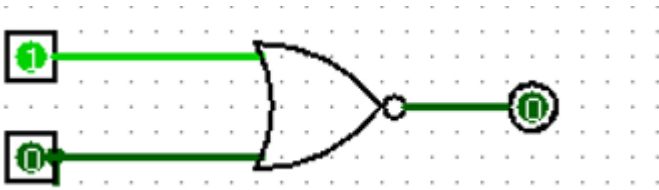
NAND gate: Outputs 0 only if *all inputs are 1* (inverse of AND).



➤ Truth table

a	b	x
0	0	1
0	1	1
1	0	1
1	1	0

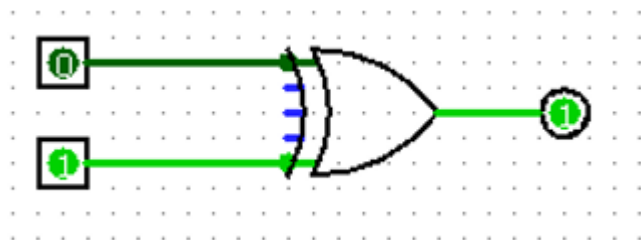
NOR gate: Outputs 1 only if *all inputs are 0* (inverse of OR).



➤ Truth table

a	b	x
0	0	1
0	1	0
1	0	0
1	1	0

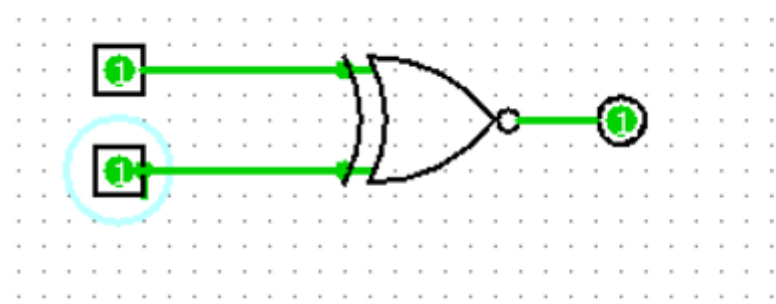
XOR gate (Exclusive OR): Outputs 1 if *inputs are different*.



➤ Truth table

a	b	x
0	0	0
0	1	1
1	0	1
1	1	0

XNOR gate (Exclusive NOR): Outputs 1 if *inputs are the same*.

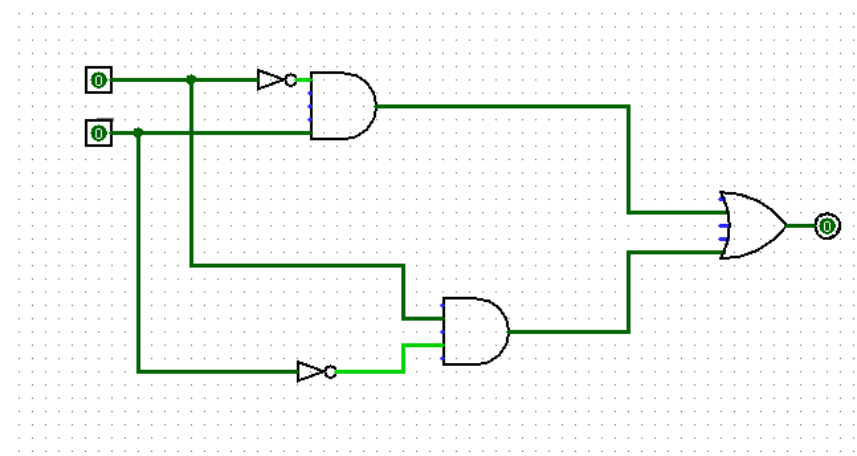


➤ Truth table

a	b	x
0	0	1
0	1	0
1	0	0
1	1	1

3) Implementation of XOR and XNOR gate using basics gates:

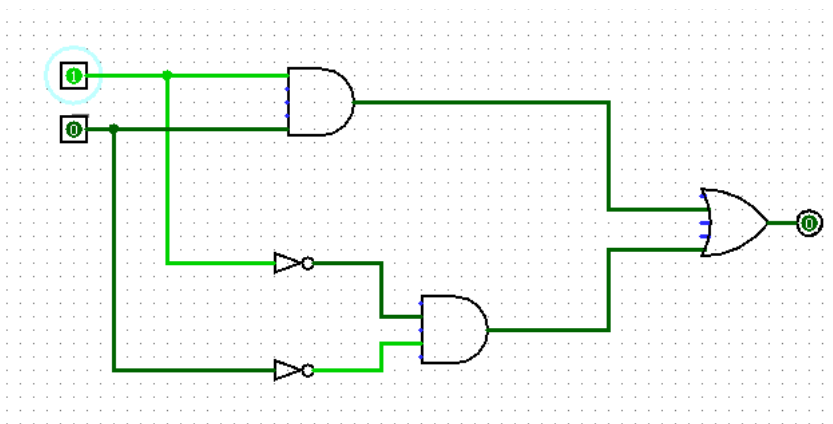
Implementation of XOR gate using basics gats.



➤ Truth table

a	b	x
0	0	0
0	1	1
1	0	1
1	1	0

Implementation of XNOR gate using basics gats.

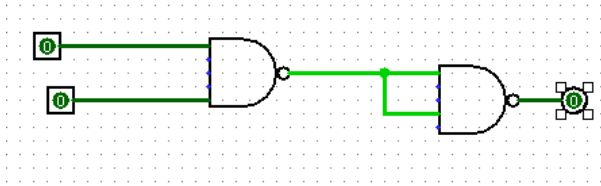


➤ **Truth table**

a	b	x
0	0	1
0	1	0
1	0	0
1	1	1

4). Universal gates implementation using NAND – Relization of AND , OR ,NOT .

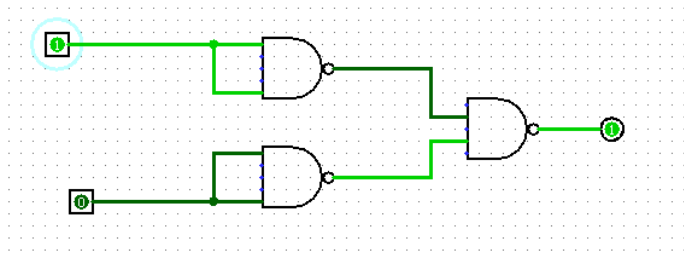
AND



➤ Truth table

a	b	x
0	0	0
0	1	0
1	0	0
1	1	1

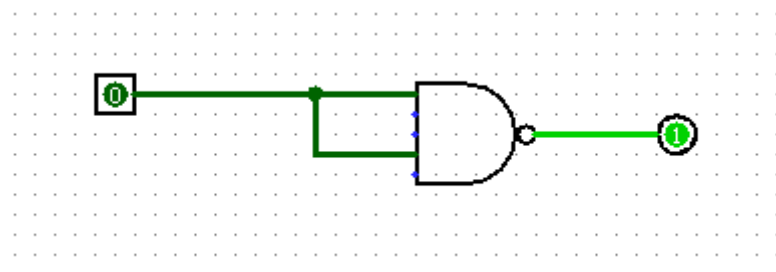
OR



➤ Truth table

a	b	x
0	0	0
0	1	1
1	0	1
1	1	1

NOT

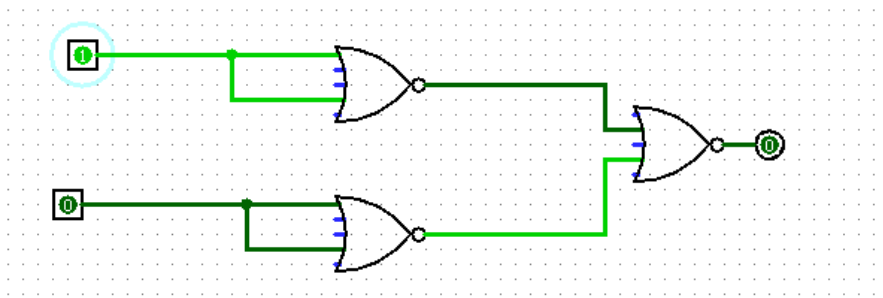


➤ Truth table

a	x
0	1
1	0

5). Universal gates implementation using NOR – Realization of AND,OR,NOT.

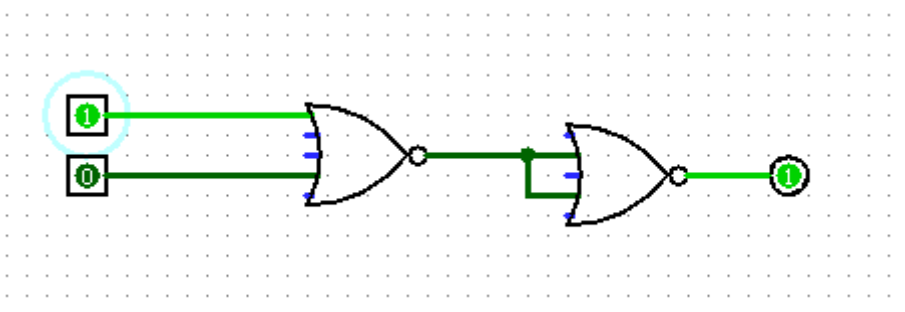
AND



➤ Truth table

a	b	x
0	0	0
0	1	0
1	0	0
1	1	1

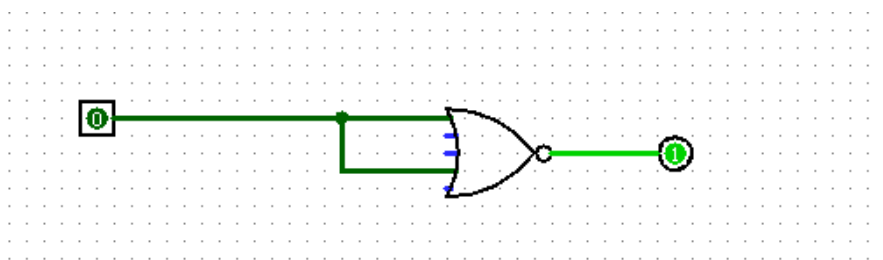
OR



➤ Truth table

a	b	x
0	0	0
0	1	1
1	0	1
1	1	1

NOT

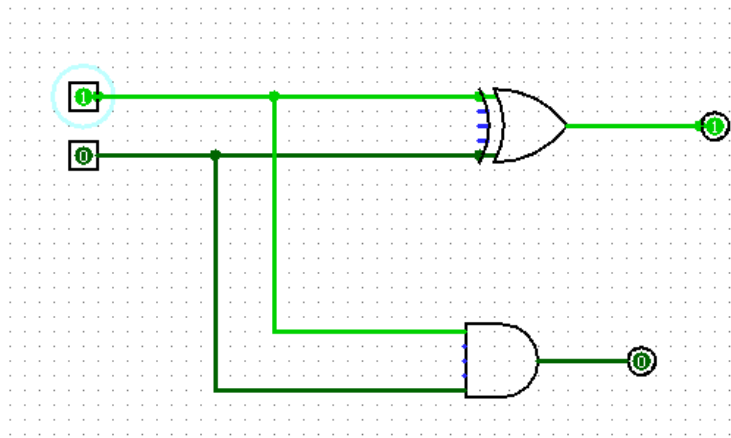


➤ Truth table

a	x
0	1
1	0

6). Design and verification of Adders – Half Adder & Full Adder.

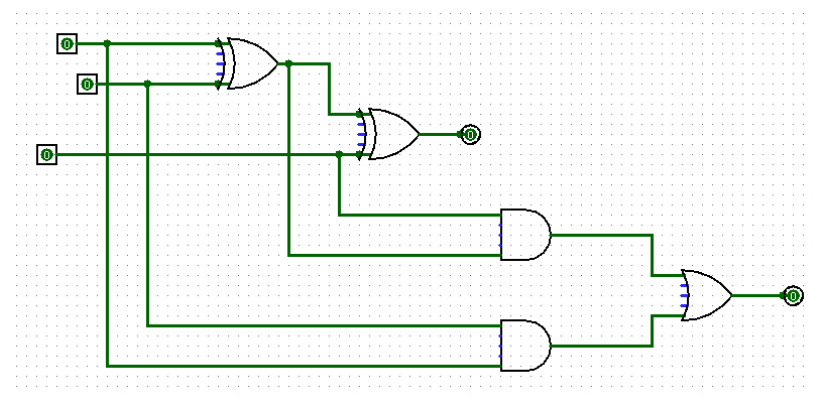
Half Adder



➤ Truth table

a	b	x	y
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Full Adder

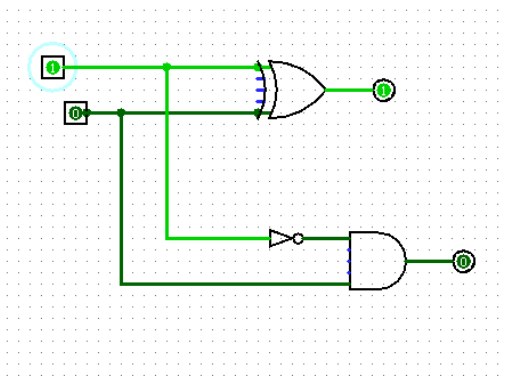


➤ Truth table

a	b	c	x	y
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

7). Design and verification of SUBTRACTOR - HALF SUBTRACTOR & FULL SUBTRACTOR .

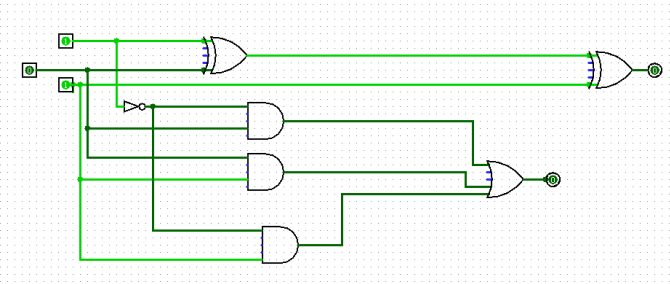
HALF SUBTRACTOR.



➤ Truth table

a	b	x	y
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

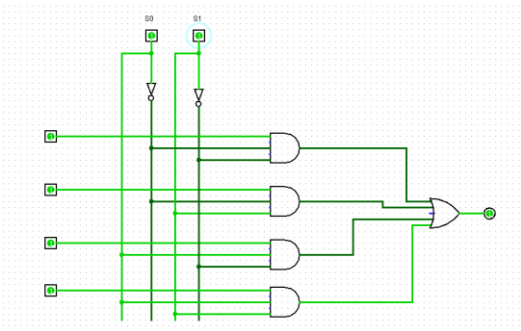
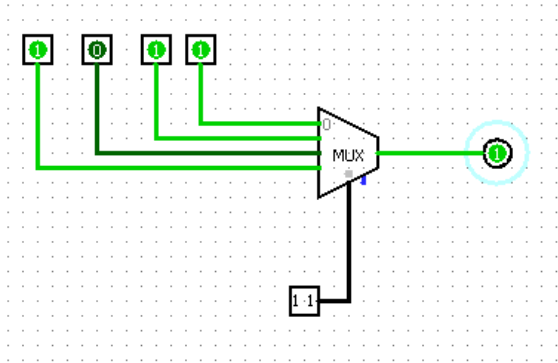
FULL SUBTRACTOR



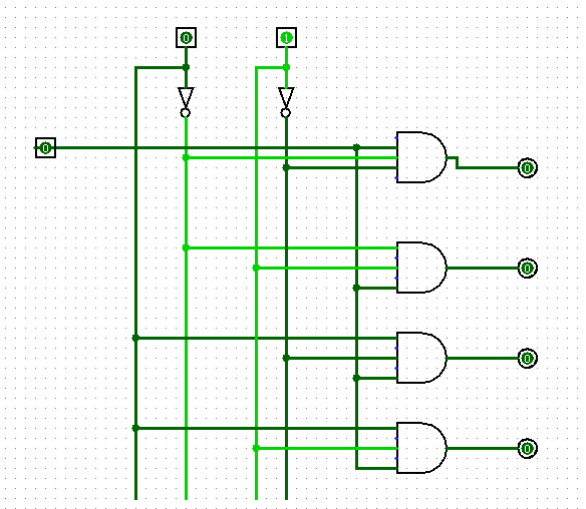
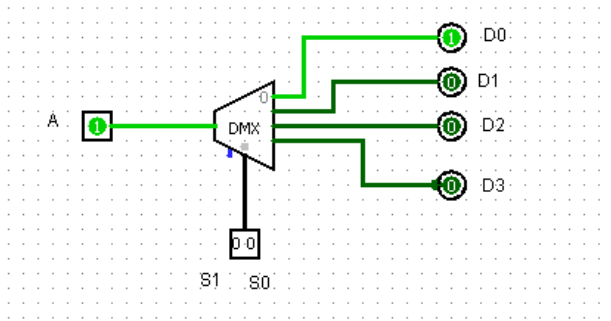
➤ Truth table

a	b	c	x	y
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

8). DESIGN AND IMPLEMENTATION OF MULTIPLEXERS – 4:1 MUX.

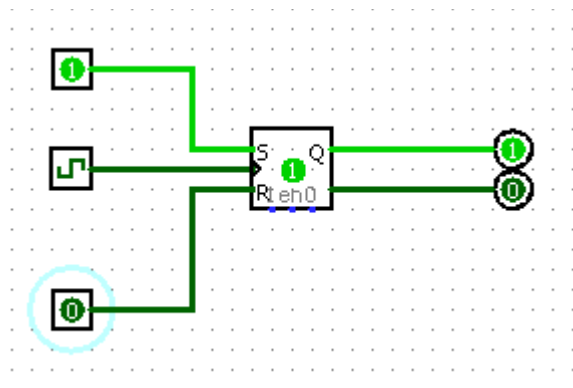


9). Design and implementation of Demultiplexers – 1:4 DEMUX.



10). DESIGN AND VERIFICATION OF FLIP – FLOP - SR FLIP-FLOP & JK FLIP-FLOP.

SR FLIP-FLOP.



➤ Truth table

a	b	x	y
0	0	0	1
0	1	0	1
1	0	0	1
1	1	0	1

JK FLIP-FLOP.

